

THE FEDERAL POLYTECHNIC, MUBI
DEPARTMENT OF COMPUTER SCIENCE TECHNOLOGY

COM 221

BASIC COMPUTER NETWORKING

COURSE UNIT: 3

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1.0 COMPUTER NETWORK

Computer Networks are used for data communications

1.1 Definition:

A computer network can be defined as a collection of nodes. A node can be any device capable of transmitting or receiving data.

The communicating nodes have to be connected by communication links.

A Computer network must be able to meet a certain criteria. The most important of these are:

- **Reliability** of the data communication process
- **Security** of the data
- **performance** by achieving higher throughput and smaller delay times

Performance can be measured in many ways, including transit time and response time.

Transit time is the amount of time required for a message to travel from one device to another.

Response time is the elapsed time between an inquiry and a response.

The performance of a network depends on a number of factors, including

- i. The number of users,
- ii. The type of transmission medium,
- iii. The capabilities of the connected hardware, and
- iv. The efficiency of the software.

Performance is often evaluated by two networking metrics: Throughput and delay.

We often need more throughput and less delay. However, these two criteria are often contradictory.

If we try to send more data to the network, we may increase throughput but we increase the delay because of traffic congestion in the network.

Reliability:

In addition to accuracy of delivery, network reliability is measured by the frequency of failure, the time it takes a link to recover from a failure, and the network's robustness in a catastrophe.

Security:

Network security issues include

- i. protecting data from unauthorized access,
- ii. protecting data from damage and
- iii. Development, and implementing policies and procedures for recovery from breaches and data losses.

1.1.1 Categories of Network

Networks are categorized on the basis of their size. The three basic categories of computer networks are:

A. Local Area Networks (LAN) is usually limited to a few kilometers of area. It may be privately owned and could be a network inside an office on one of the floor of a building or a LAN could be a network consisting of the computers in a entire building.

B. Wide Area Network (WAN) is made of all the networks in a (geographically) large area. The network in the entire state of Maharashtra could be a WAN

C. Metropolitan Area Network (MAN) is of size between LAN & WAN. It is larger than LAN but smaller than WAN. It may comprise the entire network in a city like Mumbai.

Frame Relay

Frame Relay is a technology for transmitting data between LANs or endpoints of a WAN. It specifies the physical and data-link layers of digital telecommunications channels using a packet switching methodology.

Frame Relay packages data in frames and sends it through a shared Frame Relay network. Each frame contains all necessary information for routing it to its destination. Frame Relay's original purpose was to transport data across telecom carriers' ISDN infrastructure, but it's used today in many other networking contexts.

Wide Area Network examples

The best example of a Wide Area Network is the Internet itself. Other smaller examples of WANs are:

- A network of bank cash dispensers;
- A Company network with several branch offices geographically distant.
- A school network is usually a LAN.
- LANs are often connected to WANs, for example, a school network could be connected to the Internet.
- WANs can be connected together using the Internet, leased lines or satellite links.

Wide Area Networks disadvantages

- i. WAN networks are much more expensive than home or corporate intranets.
- ii. WANs that cross international and other territorial boundaries fall under different legal jurisdictions. Disputes can arise between governments over ownership rights and network usage restrictions.

- iii. Global WANs require the use of undersea network cables to communicate across continents. Undersea cables are subject to sabotage and also unintentional breaks from ships and weather conditions. Compared to underground landlines, undersea cables tend to take much longer and cost much more to repair.

Very Fast Wide Area Network (400 Gbit Ethernet)

AT&T conducted trials in 2017 for business use of 400 gigabit Ethernet. Researchers Robert Maher, Alex Alvarado, Domaniç Lavery, and Polina Bayvel of University College London were able to increase networking speeds to 1.125 terabits per second. Christos Santis, graduate student Scott Steger, Amnon Yariv, Martin and Eileen Summerfield developed a new laser that quadruples transfer speeds with fiber optics.

If these two technologies were combined, then a transfer speed of up to 4.5 terabits per second could potentially be achieved.

❖ *Types of connection*

- A network is two or more devices connected through links.
- A link is a communications pathway that transfers data from one device to another.
 - For visualization purposes, it is simplest to imagine any link as a line drawn between two points.
- For communication to occur, two devices must be connected in some way to the same link at the same time.
 - There are two possible types of connections: point-to-point and multipoint.

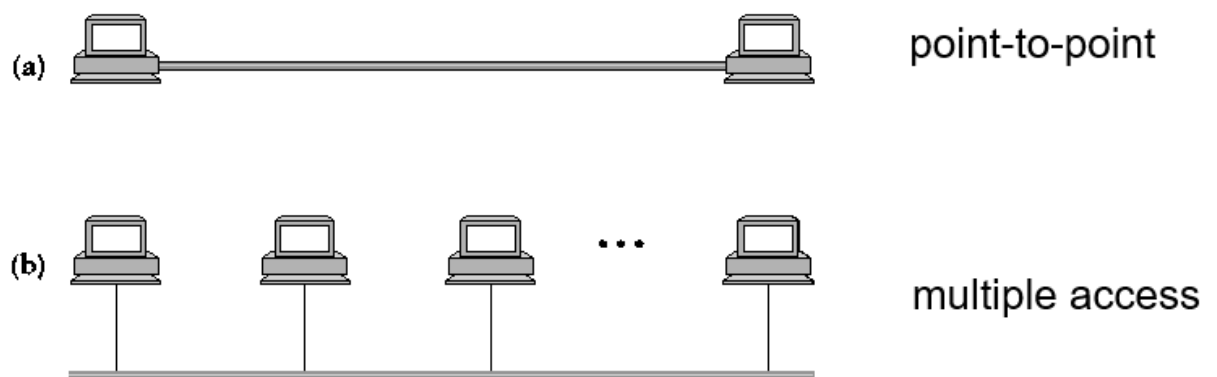
❖ *Point-to-Point*

- A point-to-point connection provides a dedicated link between two devices.
- The entire capacity of the link is reserved for transmission between those two devices.
- Most point-to-point connections use an actual length of wire or cable to connect the two ends, but other options, such as microwave or satellite links, are also possible.
 - E.g. When you change television channels by infrared remote control, you are establishing a point-to-point connection between the remote control and the television's control system.

❖ *Multipoint*

- A multipoint (also called multidrop) connection is one in which more than two specific devices share a single link.

- In a multipoint environment, the capacity of the channel is shared, either spatially or temporally.
 - If several devices can use the link simultaneously, it is a *spatially shared connection*.
 - If users must take turns, it is a timeshared connection.



Internetworking

Internetworking is combined of 2 words, inter and networking which implies an association between totally different nodes or segments. This connection area unit is established through intercessor devices akin to routers or gateway. The first term for associate degree internetwork was catenet. This interconnection is often among or between public, private, commercial, industrial, or governmental networks. Thus, associate degree internetwork could be an assortment of individual networks, connected by intermediate networking devices, that function as one giant network. Internetworking refers to the trade, products, and procedures that meet the challenge of making and administering internet works.

To enable communication, every individual network node or phase is designed with a similar protocol or communication logic, that is Transfer Control Protocol (TCP) or Internet Protocol (IP). Once a network communicates with another network having constant communication procedures, it's called Internetworking. Internetworking was designed to resolve the matter of delivering a packet of information through many links.

There is a minute difference between extending the network and Internetworking. Merely exploitation of either a switch or a hub to attach 2 local area networks is an extension of LAN whereas connecting them via the router is an associate degree example of Internetworking. Internetworking is enforced in Layer three (Network Layer) of the OSI-ISO model. The foremost notable example of internetworking is the Internet.

There is chiefly 3 units of Internetworking:

1. Extranet
2. Intranet
3. Internet

Intranets and extranets might or might not have connections to the net. If there is a connection to the net, the computer network or extranet area unit is usually shielded from being accessed from the net if it is not authorized. The net isn't thought-about to be a section of the computer network or extranet, though it should function as a portal for access to parts of the associate degree extranet.

1. **Extranet** – It's a network of the internetwork that's restricted in scope to one organization or entity however that additionally has restricted connections to the networks of one or a lot of different sometimes, however not essential. It's the very lowest level of Internetworking, usually enforced in an exceedingly personal area. Associate degree extranet may additionally be classified as a Man, WAN, or different form of network however it cannot encompass one local area network i.e. it should have a minimum of one reference to associate degree external network.
2. **Intranet** – This associate degree computer network could be a set of interconnected networks, which exploits the Internet Protocol and uses IP-based tools akin to web browsers and FTP tools that are underneath the management of one body entity. That body entity closes the computer network to the remainder of the planet and permits solely specific users. Most typically, this network is the internal network of a corporation or different enterprise. An outsized computer network can usually have its own internet server to supply users with browseable data.
3. **Internet** – A selected Internetworking, consisting of a worldwide interconnection of governmental, academic, public, and personal networks based mostly upon the Advanced analysis comes Agency Network (ARPANET) developed by ARPA of the U.S. Department of Defense additionally home to the World Wide Web (WWW) and cited as the 'Internet' to differentiate from all different generic Internetworks. Participants within the web, or their service suppliers, use IP Addresses obtained from address registries that manage assignments.

Internetworking has evolved as an answer to a few key problems: isolated LANs, duplication of resources, and an absence of network management. Isolated LANs created transmission problems between totally different offices or departments. Duplication of resources meant that constant hardware and code had to be provided to every workplace or department, as did a separate support employee. This lack of network management meant that no centralized methodology of managing and troubleshooting networks existed.

One more form of the interconnection of networks usually happens among enterprises at the Link Layer of the networking model, i.e. at the hardware-centric layer below the amount of the TCP/IP logical interfaces. Such interconnection is accomplished through network bridges and network switches. This can be typically incorrectly termed internetworking, however, the ensuing system is just a bigger, single subnetwork, and no internetworking protocol, akin to web Protocol, is needed to traverse these devices.

However, one electronic network is also reborn into associate degree internetwork by dividing the network into phases and logically dividing the segment traffic with routers. The Internet Protocol is meant to supply an associate degree unreliable packet service across the network. The design avoids intermediate network components maintaining any state of the network. Instead, this task is allotted to the endpoints of every communication session. To transfer information correctly, applications should utilize associate degree applicable Transport Layer protocol, akin to Transmission management Protocol (TCP), that provides a reliable stream. Some applications use a less complicated, connection-less transport protocol, User Datagram Protocol (UDP), for tasks that don't need reliable delivery of information or that need period of time service, akin to video streaming or voice chat.

Internetwork Addressing

Internetwork addresses establish devices severally or as members of a bunch. Addressing schemes differ based on the protocol family and therefore the OSI layer. Three kinds of internetwork addresses area units are ordinarily used: data-link layer addresses, Media Access control (MAC) addresses, and network-layer addresses.

1. **Data Link Layer addresses:** A data-link layer address unambiguously identifies every physical network association of a network device. Data-link addresses typically area units cited as physical or hardware addresses. Data-link addresses sometimes exist among a flat address area and have a pre-established and usually fastened relationship to a selected device. End

systems usually have just one physical network association, and therefore have just one data-link address. Routers and different internetworking devices usually have multiple physical network connections and so eventually have multiple data-link addresses.

2. **MAC Addresses:** Media Access management (MAC) addresses encompass a set of data-link layer addresses. MAC addresses establish network entities in LANs that implement the IEEE MAC addresses of the data-link layer. MAC addresses different area units distinctively for every local area network interface. MAC addresses are forty-eight bits long and are expressed in form of twelve hexadecimal digits. The primary half dozen hexadecimal digits, which are usually administered by the IEEE, establish the manufacturer or merchant and therefore comprise the Organizational Unique Identifier (OUI). The last half dozen positional notation digits comprise the interface serial variety or another price administered by the particular merchant. MAC addresses are typically area units referred to as burned-in addresses (BIAs) as a result of being burned into read-only memory (ROM) and are traced into random-access memory (RAM) once the interface card initializes.
3. **Network-Layer Addresses:** Network addresses sometimes exist among a gradable address area and typically area units referred to as virtual or logical addresses. The connection between a network address and a tool is logical and unfixed, it usually relies either on physical network characteristics or on groupings that don't have any physical basis. Finish systems need one network-layer address for every network-layer protocol they support. Routers and different Internetworking devices need one network-layer address per physical network association for every network-layer protocol supported.

Challenges to Internetworking

Implementing useful internetwork isn't at any certainty. There are several challenging fields, particularly in the areas of dependableness, connectivity, network management, and adaptability, and each and every space is essential in establishing associate degree economical and effective internetwork. A few of them are:-

- The initial challenge lies when we are trying to connect numerous systems to support communication between disparate technologies. For example, Totally different sites might use different kinds of media, or they could operate at variable speeds.
- Another essential thought is reliable service that should be maintained in an internetwork. Individual users and whole organizations depend upon consistent, reliable access to network resources.

- Network management should give centralized support associate degreed troubleshooting capabilities in an internetwork. Configuration, security, performance, and different problems should be adequately addressed for the internetwork to perform swimmingly.
- Flexibility, the ultimate concern, is important for network enlargement and new applications and services, among different factors.

Peer-to-peer client/server network: Here, all the computers share their resources, such as hard drives, printers and so on with all the other computers on the network. Individual resources like disk drives, CD-ROM drives, and even printers are transformed into shared, collective resources that are accessible from every PC. Unlike client-server networks, where network information is stored on a centralized file server PC and made available to tens, hundreds, or thousands client PCs, the information stored across peer-to-peer networks is uniquely decentralized. Because peer-to-peer PCs have their own hard disk drives that are accessible by all computers, each PC acts as both a client (information requestor) and a server (information provider). The peer-to-peer network is an appropriate choice when there are fewer than 10 users on the network, security is not an issue and all the users are located in the same general area.

The advantages of peer-to-peer over client-server NOSs include:

- ☐ No need for a network administrator
- ☐ Network is fast/inexpensive to setup & maintain
- ☐ Each PC can make backup copies of its data to other PCs for security.
- ☐ Easiest type of network to build, peer-to-peer is perfect for both home and office use.

Dedicated client/server network: Here, one computer is designated as server and the rest of the computers are clients. Dedicated Server Architecture can improve the efficiency of client server systems by using one server for each application that exists within an organization. The designated servers store all the networks shared files and applications programs and function only as servers and are not used as a client or workstation. Client computers can access the servers and have shared files transferred to them over the transmission medium. In some client/server networks, client computers submit jobs to one of the servers and once they process the jobs, the results are sent back to the client computer. In general, the dedicated client/server model is preferable to the peer-to-peer client/server model for general purpose data networks.

1.1.2 PROTOCOL

A Protocol is one of the components of a data communications system. Without protocol communication cannot occur. The sending device cannot just send the data and expect the receiving device to receive and further interpret it correctly.

When the sender sends a message it may consist of text, number, images, etc. which are converted into bits and grouped into blocks to be transmitted and often certain additional information called control information is also added to help the receiver interpret the data.

For successful communication to occur, the sender and receiver must agree upon certain rules called protocol.

A Protocol is defined as a set of rules that governs data communications.

A protocol defines what is to be communicated, how it is to be communicated and when it is to be communicated.

1.1.3 Elements of a Protocol

There are three key elements of a protocol:

A. Syntax

- It means the structure or format of the data.
- It is the arrangement of data in a particular order.

B. Semantics

- It tells the meaning of each section of bits and indicates the interpretation of each section.
- It also tells what action/decision is to be taken based on the interpretation.

C. Timing

- It tells the sender about the readiness of the receiver to receive the data
- It tells the sender at what rate the data should be sent to the receiver to avoid overwhelming the receiver.

1.1.4 STANDARDS IN NETWORKING

Standards are necessary in networking to ensure interconnectivity and interoperability between various networking hardware and software components.

Without standards we would have proprietary products creating isolated islands of users which cannot interconnect.

1.1.5 Concept of Standard

Standards provide guidelines to product manufacturers and vendors to ensure national and international interconnectivity.

Data communications standards are classified into two categories:

1. De facto Standard

These are the standards that have been traditionally used and mean **by fact** or **by convention**

These standards are not approved by any organized body but are adopted by widespread use.

2. De jure standard

It means by **law** or **by regulation**.

These standards are legislated and approved by a body that is officially recognized.

Standard Organizations in field of Networking

Standards are created by standards creation committees, forums, and government regulatory agencies.

Examples of Standard Creation Committees:

1. International Organization for Standardization (ISO)
2. International Telecommunications Union Telecommunications Standard (ITU-T)
3. American National Standards Institute (ANSI)
4. Institute of Electrical & Electronics Engineers (IEEE)
5. Electronic Industries Associates (EIA)

Examples of Forums

1. ATM Forum (Asynchronous Transfer Mode technology)
2. MPLS Forum (Multi-Protocol Label Switching)
3. Frame Relay Forum

Examples of Regulatory Agencies:

1. Federal Communications Committee (FCC)

UTP CATEGORIES

Category 1	Voice only (Telephone)
Category 2	Data to 4 Mbps (Localtalk)
Category 3	Data to 10Mbps (Ethernet)
Category 4	Data to 20Mbps (Token ring)
Category 5	Data to 100Mbps (Fast Ethernet)
Category 5e	Data to 1000Mbps (Gigabit Ethernet)
Category 6	Data to 2500Mbps (Gigabit Ethernet)

A media access method refers to the manner in which a computer gains and controls access to the network's physical medium (e.g., cable).

MEDIA ACCESS METHODS

Common media access methods include the following:

CSMA/CD

CSMA/CA

Token Passing

One of the primary concerns with media access is how to prevent packets from colliding when two or more computers try to transmit simultaneously.

Each of the methods listed above takes a different approach to this problem.

Data transmitted over a network is sent one bit at a time.

A bit is either a 1 or a 0 represented by a voltage change (on or off) or a light pulse.

If two stations are transmitting at the same time, it is possible that the signals may overlap, resulting in garbled data.

Such overlapping is referred to as a "collision."

CSMA/CD stands for Carrier-Sense Multiple Access with Collision Detection.

It is a media access method which means it defines how the network places data on the cable and how it takes it off.

CSMA/CD specifies how bus topologies such as Ethernet handle transmission collisions. A collision occurs when two or more computers transmit signals at the same time.

"Carrier Sense" means that each station on the LAN continually listens to (tests) the cable for the presence of a signal prior to transmitting.

"Multiple Access" means that there are many computers attempting to transmit and compete for the opportunity to send data (i.e., they are in contention).

"Collision Detection" means that when a collision is detected, the stations will stop transmitting and wait a random length of time before retransmitting.

CSMA/CD works best in an environment where relatively fewer, longer data frames are transmitted. This is in contrast to token passing which works best with a relatively large amount of short data frames.

Because CSMA/CD works to control or manage collisions rather than prevent them, network performance can be degraded with heavy traffic.

The greater the traffic, the greater the number of collisions and retransmissions.

CSMA/CD is used on Ethernet networks.

Operation:

In its most simple form it operates as follows:

A station that wishes to transmit on the network checks to see if the cable is free.

If the cable is free, the station starts transmitting.

However, another station may have detected a free cable at the same instant and also start transmitting. The result is a "collision."

Once the collision is detected, all stations immediately stop transmitting.

Stations then wait a random length of time before checking the cable and then retransmit

CSMA/CA stands for Carrier-Sense Multiple Access with Collision Avoidance and is a media access method very similar to CSMA/CD.

The difference is that the CD (collision detection) is changed to CA (collision avoidance).

Instead of detecting and reacting to COLLISIONS, CSMA/CA tries to avoid them by having each computer signal its intention to transmit before actually transmitting.

In effect, the transmitting computer gives a 'heads up' prior to transmitting.

Although CSMA/CA can prevent collisions, it comes with a cost in the form of the additional overhead incurred by having each workstation broadcast its intention prior to transmitting.

Thus, CSMA/CA is slower than CSMA/CD.

CSMA/CA is used on Apple networks and on WiFi (IEEE 802.11) networks.

Token passing is a media access method by which collisions are prevented.

Collisions are eliminated under token passing because only a computer that possesses a free token (a small data frame) is allowed to transmit.

The token passing method also allows different priorities to be assigned to different stations on the ring.

Transmissions from a station with higher priority take precedence over stations with lower priority.

Token passing works best in an environment where a relatively large number of shorter data frames are being transmitted.

(As opposed to CSMA/CD which works best in an environment where relatively fewer, longer data frames are being transmitted.)

Token passing is used on Token Ring networks

In its most simple form it operates as follows:

A station that wishes to transmit on the network waits until it will receive a free token.

The sending station transmits its data with the token.

The data travels to the recipient without stopping at other stations (it is just relayed).

The receiving station receives the data and returns the token to the sender as an acknowledgment.

The sender receives acknowledgment and releases the token to next station.

The token continues being passed along the ring until it is "seized" by the next station that wants to transmit.

Ethernet is a popular, relatively inexpensive, easy-to-install LAN architecture with the following characteristics:

Uses the CSMA/CD media access control.

Data transmission normally occurs at 100 Mbps (10Mbps in the early forms and 10Gbps in the most recent forms).

Typically implemented in a star topology (early versions used bus topology as well).

Ethernet LANs are normally distinguished by the type of cable they use Twisted Pair (Thinnet or Thicknet were also used in the past).

The Ethernet architecture conforms to most but not all of the IEEE 802.3 specification (the physical layers are identical but the MAC layers are somewhat different).

An Ethernet LAN is often described in terms of three parameters:

Transmission rate, transmission type, and segment distance or cable type.

"100baseT" means:

100 - Transmission rate or throughput of 100Mbps

Base - transmission type is baseband rather than broadband network (i.e., the signal is placed directly on the cable, one signal at a time)

T – The cable type (e.g. Twisted pair)

Token ring is a relatively expensive LAN architecture that was strongly influenced by IBM.

It is very stable and can be expanded without a significant degradation in network performance.

Token ring uses the token passing media access control. Data transmission normally occurs at 4 or 16 Mbps depending on the cable.

Token ring is normally implemented in a logical ring/physical star topology with a MAU (Multistation Access Unit) as the hub.

The maximum number of stations on one ring is 260 for shielded twisted pair and 72 for unshielded twisted pair (UTP).

There can be up to 33 MAUs per ring.

Token Ring LANs normally use shielded twisted pair (STP) but may also use unshielded twisted pair (UTP) or fiber-optic cable.

The maximum distance to the MAU from the workstation depends on the cable and varies from 45 meters for UTP to 100 meters for STP.

The Token Ring architecture conforms generally to the IEEE's 802.5 specification

Few types of Ethernet: 10Base2, 10Base5, 10BaseT and 10BaseF, 100BaseT, 100BaseF, etc..

Ethernet is generally less expensive and easier to install than Token Ring.

Token Ring is generally more secure and more stable than Ethernet, but not used anymore in typical LAN configurations.

It is usually more difficult to add more computers on a Token Ring LAN than it is to an Ethernet LAN.

However, as additional computers are added, performance degradation will be less pronounced on the Token Ring LAN than it will be on the Ethernet LAN.

Ethernet uses CSMA/CD media access control and Token Ring uses token passing.

This makes Ethernet better suited in a situation where there are a large number of computers sending fewer, larger data frames.

Token Ring is better suited for small to medium size LANs sending many, smaller data frames.

Networking devices

There are different types of electronic devices are used in networking which are known as network devices or network equipment. In a computer network, network devices are mainly used for transmitting and receiving the data quickly and securely in between computers, fax machines, printers, etc. These devices may be intra network or internetwork. There are some devices are installed on the device such as RJ45 connector otherwise NIC card, whereas some devices are part of the network namely switch, router, etc. These devices are specific devices, handles digital or electrical connections to perform their exclusive roles very efficiently.

Definition: The devices which are used for communication between different hardware's used in the computer network are known as network devices. These devices are also known as physical devices, networking hardware, and network equipment otherwise computer networking devices. In a computer network, each network device plays a key role based on their functionality, and also works for different purposes at different segments.

Types of Network Devices

There are different types of network devices used in a computer network which include the following.

- Network Hub
- Network Switch

- Modem
- Network Router
- Bridge
- Repeater

Network Hub

The network hub is one kind of networking device in a computer network, used to communicate with various network hosts and also for data transferring. The transferring of data in a computer network can be done in the form of packets. Whenever the data processing can be done from a host to a network hub, then the data can transmit to all the connected ports. Similarly, all the ports identify the data path which leads to inefficiencies & wastage. Because of this working, a network hub cannot be so safe and secure. In addition, copying the data packets on all the ports will make the hub slower which leads to the utilize of the network switch.



Network hubs are classified into two types like active hub & passive hub.

Active Hub

These hubs have their own power supply and these hubs are used to clean, increase & transmit the signal using the network. It works as a wiring center & repeater. Active hubs play a key role in extending the distance between nodes.

Passive Hub

These hubs collect wiring from the power supply and different nodes of an active hub. These hubs transmit the signals over the network without improving & cleaning them. These hubs are not suitable for extending the distance between nodes like an active hub.

Network Switch

Similar to a hub, this is also working at the layer in the LAN and a switch is more clever compare with a hub. As the hub is used for data transferring, whereas a switch is used for filtering & forwarding the data. So this is the more clever technique to deal with the data packets.

Whenever a data packet is obtained from the interfaces in the switch, then the data packet can be filtered & transmits to the interface of the proposed receiver. Due to this reason, a switch maintains a content addressable memory table to maintain system configuration as well as memory. This table is also named as FIB (forwarding information base) otherwise forwarding table.



Modem

A modem is the most important network device and it is used daily in our life. If we notice the internet connection to homes was given with the help of a wire. then wire carries internet data from one place to another. But, every computer gives digital or binary data in the form of zeros & ones.

The full form of the modem is a modulator and a demodulator. So it modulates as well as demodulates the signal among the computer and a telephone line because the computer generates digital data whereas the telephone line generates an analog signal.



Network Router

A network router is one kind of network device in a computer network and it is used for routing traffic from one network to another. These two networks could be private to a public company network. For example, here a router is considered as traffic police at the junction, he directs dissimilar traffic networks to dissimilar directions.



Bridge

A Bridge in the computer network is used to unite two or more network segments. The main function of a bridge in network architecture is to store as well as transmit frames among the various segments. Bridges use MAC (Media Access Control) hardware for transferring frames.

These are also used for connecting two physical local area networks to a larger logical local area network. In the OSI model, bridges work at the data link & physical layers to divide the networks from larger to smaller by controlling the data flow between the two. In recent years, bridges are replaced by switches to provide more functionality.



Repeater

The operating of a repeater can be done at the physical layer. The main function of this device is to reproduce the signal on a similar network before the signal gets weak otherwise damaged. The significant point to be noted regarding these devices is that they do not strengthen the signal. Whenever the signal gets weak, then they reproduce it at the actual strength. A repeater is a two-port device.



Gateway

Generally, a gateway performs at the session & transport layers in the OSI model. Gateways offer conversion between networking technologies like OSI (Open System Interconnection) & TCP/IP. Because of this, these are connected to two or many autonomous networks, where each network has its own domain name service, routing algorithm, topology, protocols, and procedures of network administration & policies.

Gateways execute all the functions of routers. Actually, a router with additional conversion functionality is a gateway, so the conversion between various network technologies is known as a protocol converter.



Router

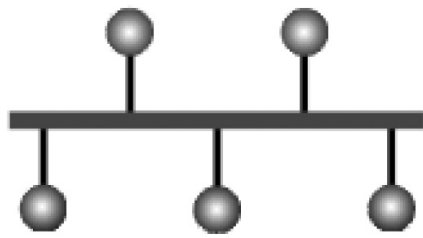
The Router is also called a bridging router and the main function of this is to combine the features of both router & bridge and router. It performs either at the network layer or the data link layer. When it works as a router, it is used for routing packets across networks whereas it works as a bridge; it is used for filtering LANs traffic.

REVIEW QUESTIONS

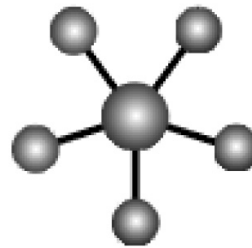
1. Differentiate between data & information. What are the different forms in which data can be represented?
2. What are the characteristics of data communication?
3. What are the components of a data communication system?
4. Define computer network and categorize.
5. Explain protocols in details

Understand the Concept of Computer Network Topology

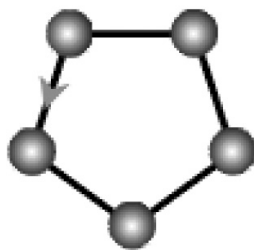
Network topology is the schematic description of a network arrangement, connecting various nodes (sender and receiver) through lines of connection. Topology is the schematic diagram showing how the networks are connected physically and logically. Physical topology is the physical layout of nodes, workstations and cables in the network; while logical topology is the way information flows between different components. So network topology is the arrangement of the various elements of a computer network. Details of network topologies are given below and are represented in Fig. 6.1



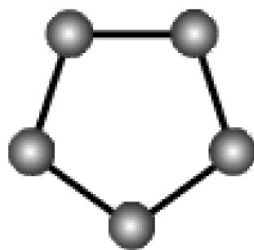
Bus



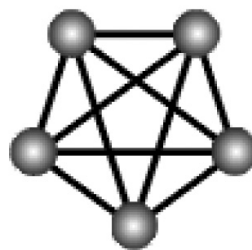
Star



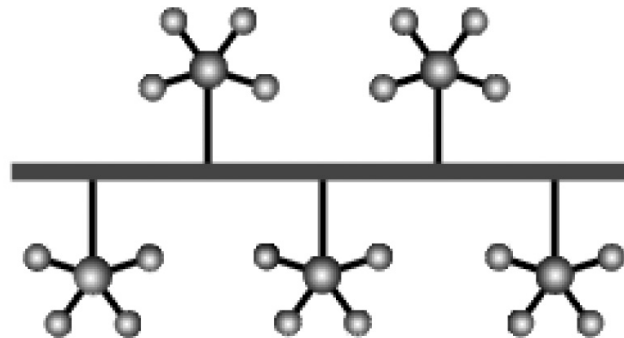
Token Ring



Ring



Mesh



Tree

1. Bus Topology: In this type of topology all computers are connected by a single length of cable with a terminator at each end. The bus topology is the simplest and most widely used local area network design. Only one computer at a time can send a message. It transmits data only in one direction and every device is connected to a single cable. So it affects the speed of the network. It is cost effective and requires least amount of cable compared to other topologies. It is normally used in smaller networks and is very easy to understand. But it is slower and if the main cable fails, the entire network fails.

2. Star Topology: The star network is frequently used to connect one or more small computers or peripheral devices to a large host computer or CPU. Each computer on a star network communicates with a central hub that re-sends the message either to all the computers or only to the destination computer.

Every node has a dedicated connection to the central hub. It is fast with few nodes and low network traffic. It is easy to troubleshoot and easy to setup. But cost of installation is high and is expensive to use as well. If the host computer fails, the entire network gets affected.

3. Token Ring: A token ring network is a Local Area Network (LAN) in which all computers are connected in a ring or star topology and a bit- or token passing scheme is used in order to prevent the collision of data between two computers that want to send messages at the same time. The token ring protocol was developed by IBM. The access method used involves token passing. In token ring, the computers are connected so that the signal travels around the network from one computer to another in a logical ring.

4. Ring Topology: In this type of topology the workstations are connected in a circle using cable segments. In this layer each node is physically connected to only two other nodes i.e., adjacent pairs of workstations are directly connected. Ring topology can function without server. The ring topology may prove optimum when system requirements are modest and workstations are at scattered points.

5. Mesh Topology: It is a type of network setup where each of the computers and network devices are interconnected with one another. A mesh topology is most often used in large backbone networks in which failure of a single switch or router can result in a large portion of a network going down. A mesh network topology is a decentralized design in which each node on the network connects to at least two other nodes. Mesh networks can use a full mesh topology or a partial mesh topology. In a full mesh topology, each network node is connected to all the other nodes in the network. In a partial mesh topology, at least one node connects directly to every other node while others may only connect to those nodes they exchange data with on a frequent basis.

6. Tree Topology: The tree topology, also known as **Hierarchical topology** uses two or more star networks connected together. It could also employ a combination of bus and star topologies. This topology divides the network into multiple levels of network. It has a root node, intermediate node, and ultimate nodes. The tree network topology is ideal when the workstations are located in groups, with each group occupying a relatively small physical region. An example is a university campus in which each building has its own star network, and all the central computers are linked in a campus-wide system. It is easy to add or remove workstations from each star network. Entire star networks can be added to, or removed from, the bus.

Connecting Media: Wired and Wireless and their characteristics

Nowadays, internet users wish to enjoy accessing the internet even when they are away from their home and office place. Wireless modem is a modem that sends or receives network signals without

the use of cable connections. It can access the internet without using any wired connectors or cables. Because of being wireless, it is much faster, reliable and is less expensive.

Today, many companies are into manufacturing wireless modems. These are designed as per the user requirements and are not expensive. Some of the famous and common wireless modems which are available include connect cards, USB sticks, Wi-Fi devices and wireless routers.

Connect Cards

It is the starting series of wireless modems which first appeared in two versions:

- PC data cards
- Connect cards

Such wireless modems are very small and compact. They are used to provide internet facility in laptops, personal computers or routers.

USB Sticks

It is a type of stick that is connected with the wireless modem and gives good internet speed in desktops and laptops. The size of USB stick is same as the size of a pen drive. The stick fits correctly into the USB port available in either the desktop or laptop. These sticks are not plug and play because it requires certain installation drivers in order to work. Such type of USB sticks are very easy to carry and can work anywhere.

Mobile Hotspots

Wireless modems serve as portable internet hotspots. Internet products such as Novatel's Mi-Fi routers get a wireless broadband network and move the signal to a particular range in the same way as Wi-Fi hotspots. It will make the respective devices with Wi-Fi features along with wireless broadband network that can be of much use to the user who use Computers, Smartphone and tablets while moving here and there.

Wireless Routers

These are routers which are especially designed for home users as there is no need of a transmission cable. This router works without the internet cable and can receive or send the signal faster than a normal router.

Wireless transmission is a form of unguided media. Wireless communication involves no physical link established between two or more devices, communicating wirelessly. Wireless signals are spread over in the air and are received and interpreted by appropriate antennas.

When an antenna is attached to electrical circuit of a computer or wireless device, it converts the digital data into wireless signals and spread all over within its frequency range. The receptor on the other end receives these signals and converts them back to digital data.

A little part of electromagnetic spectrum can be used for wireless transmission.



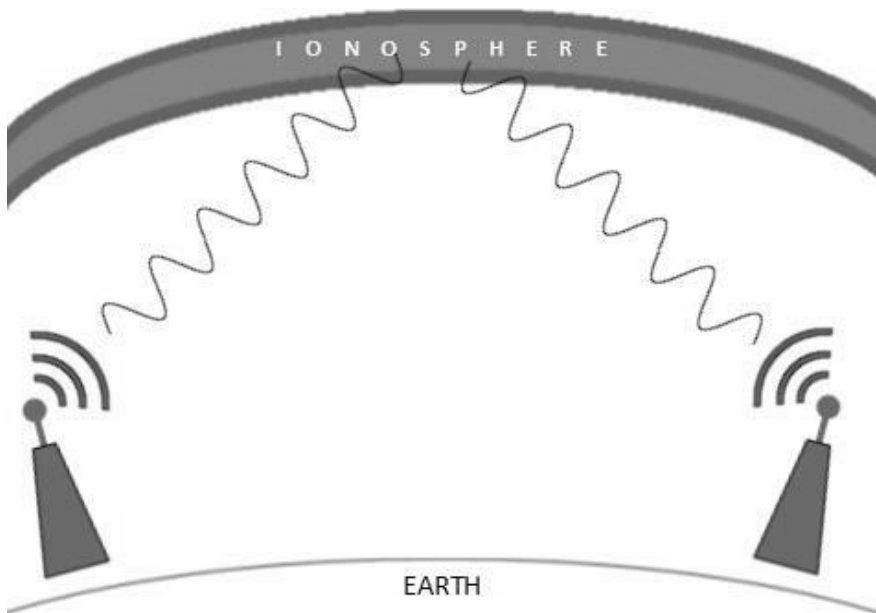
Radio Transmission

Radio frequency is easier to generate and because of its large wavelength it can penetrate through walls and structures alike. Radio waves can have wavelength from 1mm – 100,000km and have frequency ranging from 3Hz (Extremely Low Frequency) to 300 GHz (Extremely High Frequency). Radio frequencies are sub-divided into six bands.

Radio waves at lower frequencies can travel through walls whereas higher RF can travel in straight line and bounce back. The power of low frequency waves decreases sharply as they cover long distance. High frequency radio waves have more power.

Lower frequencies such as VLF, LF, MF bands can travel on the ground up to 1000 kilometers, over the earth's surface.

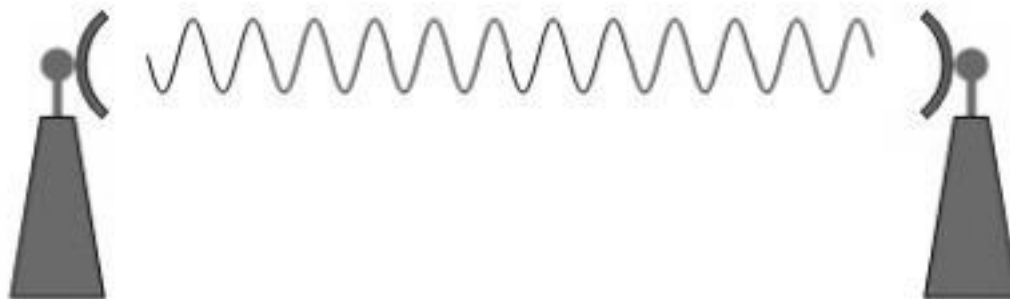
Radio waves of high frequencies are prone to be absorbed by rain and other obstacles. They use Ionosphere of earth atmosphere. High frequency radio waves such as HF and VHF bands are spread upwards. When they reach Ionosphere, they are refracted back to the earth.



Microwave Transmission

Electromagnetic waves above 100MHz tend to travel in a straight line and signals over them can be sent by beaming those waves towards one particular station. Because Microwaves travels in straight lines, both sender and receiver must be aligned to be strictly in line-of-sight.

Microwaves can have wavelength ranging from 1mm – 1meter and frequency ranging from 300MHz to 300GHz.



Microwave antennas concentrate the waves making a beam of it. As shown in picture above, multiple antennas can be aligned to reach farther. Microwaves have higher frequencies and do not penetrate wall like obstacles.

Microwave transmission depends highly upon the weather conditions and the frequency it is using.

Infrared Transmission

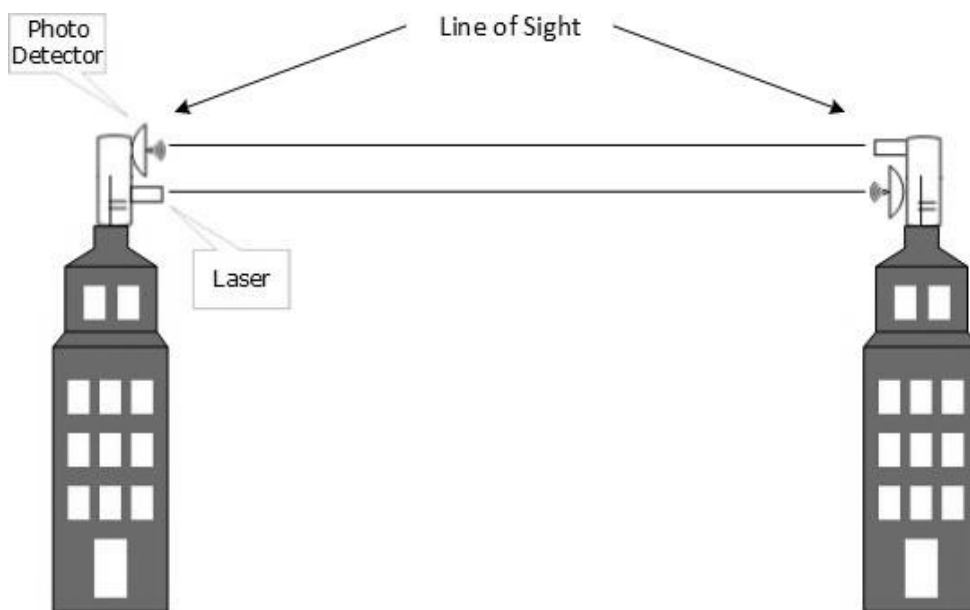
Infrared wave lies in between visible light spectrum and microwaves. It has wavelength of 700nm to 1mm and frequency ranges from 300GHz to 430THz.

Infrared wave is used for very short range communication purposes such as television and its remote. Infrared travels in a straight line hence it is directional by nature. Because of high frequency range, Infrared cannot cross wall-like obstacles.

Light Transmission

Highest most electromagnetic spectrum which can be used for data transmission is light or optical signaling. This is achieved by means of LASER.

Because of frequency light uses, it tends to travel strictly in straight line. Hence the sender and receiver must be in the line-of-sight. Because laser transmission is unidirectional, at both ends of communication the laser and the photo-detector needs to be installed. Laser beam is generally 1mm wide hence it is a work of precision to align two far receptors each pointing to lasers source.



Laser works as Tx (transmitter) and photo-detectors works as Rx (receiver).

Lasers cannot penetrate obstacles such as walls, rain, and thick fog. Additionally, laser beam is distorted by wind, atmosphere temperature, or variation in temperature in the path.

Laser is safe for data transmission as it is very difficult to tap 1mm wide laser without interrupting the communication channel.

Ethernet cables are grouped in sequence numbered categories supported different specifications; typically the category is updated with further clarification or testing standards. These categories are how we can simply know what type of cable we need for a particular application.

It is noticed that Category 5 cable was revised and upgraded with category 5 enhanced (Cat-5e) cables with similar physical look inside and outside the cable, with required testing standards.

Category 6 was revised with greater than before category 6 (Cat-6a) which shows testing for 500 MHz communication. It is noted that higher communication frequency will eliminate alien crosstalk (AXT) that allow for longer range upto 10

Gb/s.

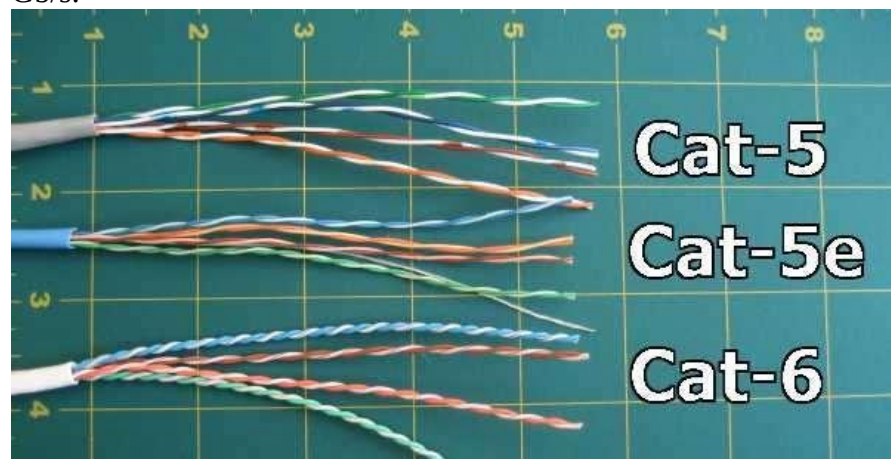


Fig 1.1 Types of CAT cable

It is found that certain Cat-6 cables will carry out nylon spline that will help to remove crosstalk. It is noted that mostly spline is not required in Cat-5 cable. In Cat-6 cable, spline is not required either the cable is long or short and as per standard. Fig 1.1 shows Cat-5e cable with spline.

Generally, Ethernet cables are twisted pair where manufactures uses shielding so as to safeguard from interference, while unshielded twisted pair also applied for cables among computer and wall which uses shielding cable for areas having high interference and running cables outdoors or inside walls.



Fig. UTP and STP Cable

There are different ways to shield an Ethernet cable, but typically it involves putting a shield around each pair of wire in the cable. This protects the pairs from crosstalk internally. It is found that manufactures protect the cables from foreign identities simply by screening UTP or STP cables.

<i>Category</i>	<i>Specification</i>	<i>Data Rate (Mbps)</i>	<i>Use</i>
1	Unshielded twisted-pair used in telephone	< 0.1	Telephone
2	Unshielded twisted-pair originally used in T-lines	2	T-1 lines
3	Improved CAT 2 used in LANs	10	LANs
4	Improved CAT 3 used in Token Ring networks	20	LANs
5	Cable wire is normally 24 AWG with a jacket and outside sheath	100	LANs
5E	An extension to category 5 that includes extra features to minimize the crosstalk and electromagnetic interference	125	LANs
6	A new category with matched components coming from the same manufacturer. The cable must be tested at a 200-Mbps data rate.	200	LANs
7	Sometimes called SSTP (shielded screen twisted-pair). Each pair is individually wrapped in a helical metallic foil followed by a metallic foil shield in addition to the outside sheath. The shield decreases the effect of crosstalk and increases the data rate.	600	LANs

Categories of unshielded twisted-pair cables

RJ-45 connectors

Creating a Switched Wired Ethernet LAN

RJ connector is registered jack (RJ) standardized physical network interface that connects telecommunications or data equipment. RJ45 could be a standard type of connector for network cables. RJ45 connectors are most commonly seen with Ethernet cables and networks. It's an eight pins to which the wire strands of a cable interface electrically. Standard RJ-45 pin outs define the arrangement of the individual wires needed when attaching connectors to a cable.

Several other kinds of connectors closely resemble RJ45 and might be easily confused for each other. The RJ-11 connectors used with telephone cables, for example, are only slightly smaller (narrower) than RJ-45 connectors.



Fig. RJ Connector

It is noted that connectors with required jacks are mostly modular connector which can be 50-pin miniature ribbon connector type. Such types are most common twisted pair connector types which is 8-position, 8-contact (8P8C) modular plug and jack called as RJ45 connector. RJ-45 conductor data cable consists of 4 pairs of wires each consists of a solid colored wire and a strip of the same colour. There are two wiring standards for RJ-45 wiring:

RJ-45 conductor data cable consists of 4 pairs of wires each consists of a solid colored wire and a strip of the same colour. There are two wiring standards for RJ-45 wiring:

• T-568A

• T-568B

There are 4 pairs of wires; 10BaseT/100BaseT Ethernet uses only 2 pairs:

• Orange

• Green

Apart from this, other two colors are:

• Blue

• Brown

This is used for second Ethernet line or for phone connections. The two wiring standards are used to create a cross-over cable:

• T-568A on one end

• T-568B on other end

Straight-through cable:

• T-568B

• T-568A

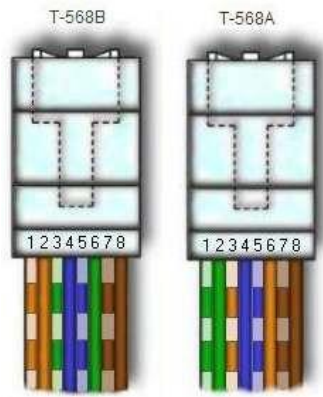


Fig. Cable T568B and T568A

The RJ45 data cable is used to connect computers to Ethernet switch by straight-through cables. The RJ45 cable uses only 2-pairs of wires:

- ✓ Orange (pins 1 & 2)
- ✓ Green (pins 3 & 6)
- ✓ Pins 4, 5 (Blue) and 7, 8 (Brown) are NOT used

Straight-through cable connects:

- ✓ pin 1 to pin 1
- ✓ pin 2 to pin 2
- ✓ pin 3 to pin 3
- ✓ pin 6 to pin 6

Cross-over cables are used to connect:

- ✓ TX+ to RX+
- ✓ TX- to RX-

That connects pin 1 to pin 3, pin 2 to pin 6, pin 3 to pin 1 and pin 6 to pin 2. The unused pins are generally connected straight-through in both straight-through and cross-over cables.

To network two computers without a hub, a cross-over cable is used. Cross-over cable is also used to connect a router to a computer, or Ethernet switch (hub) to another Ethernet switch without an uplink. Most Ethernet switches today provide an uplink port, which prevents a use of cross-over cable to daisy chain

another Ethernet switch. Straight-through cables are used to connect a computer to an Ethernet switch, or a router to an Ethernet switch.

It is noted that RJ45 cables carries 8 color coded wires, and the plugs have 8 pins and conductors. In this eight wires are used as 4 pairs, each representing positive and negative polarity.

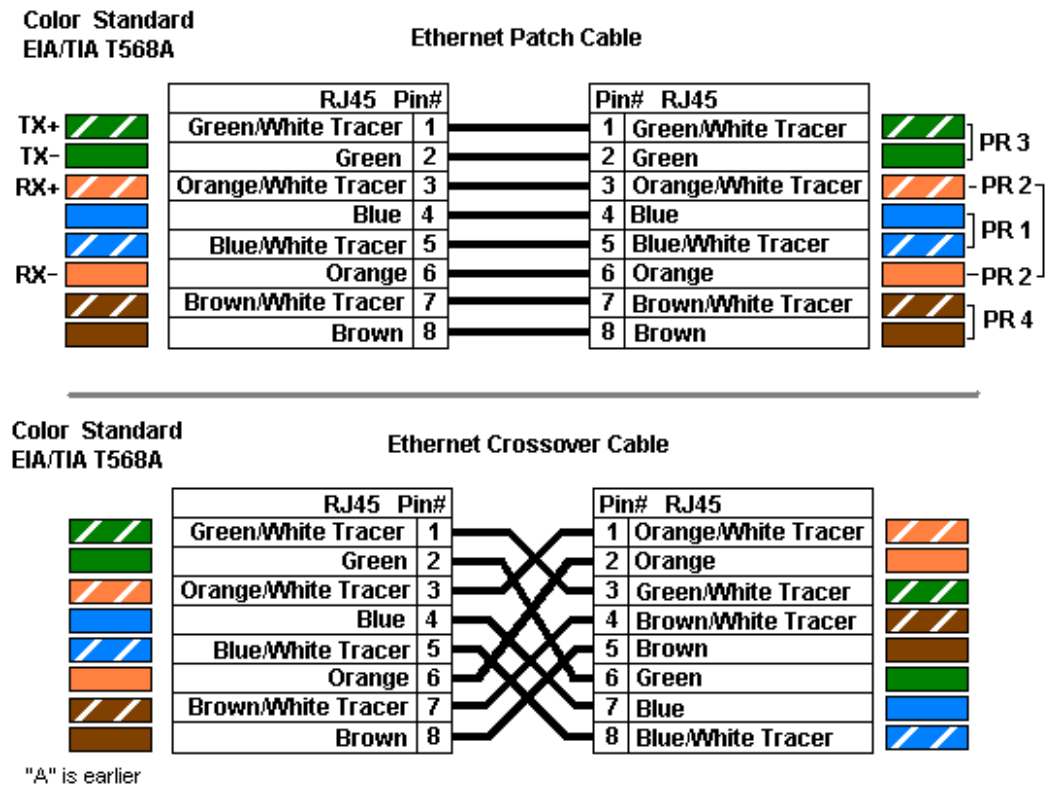


Fig. Colour Standard of T568A

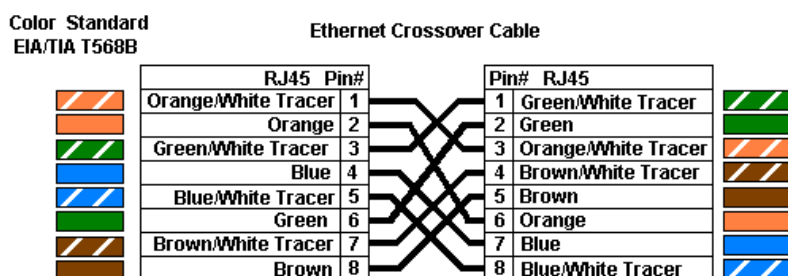
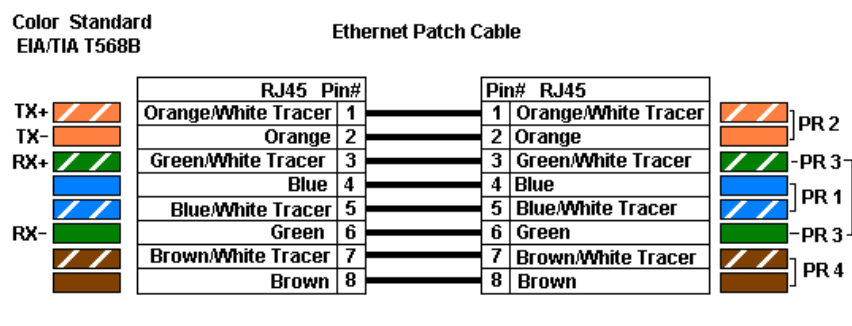


Fig. Colour Standard of T568B

The most commonly used wiring standard for 100baseT is T-586B standards. Prior to EIA 568A and 568B standards, the color-coded scheme was used to wire RJ45 cables. The table below shows pin and color schemes used in various setup.

Table: Pin and Colour Scheme of RJ45 Connector

Pin	Colored Scheme	T-568B (Common)	T-568A
1	Blue	Orange Stripe	Green Stripe
2	Orange	Orange	Green
3	Black	Green Stripe	Orange Stripe
4	Red	Blue	Blue
5	Green	Blue Stripe	Blue Stripe
6	Yellow	Green	Orange
7	Brown	Brown Stripe	Brown Stripe
8	White (or Grey)	Brown	Brown

Creating a Switched Wired Ethernet LAN

Crimping a UTP cable to RJ-45 connector

A UTP cable is one of the most popular LAN cables which consist of 4 twisted pairs of metal wires. Adding RJ45 connectors at both the ends of the UTP cable will allow the cable to work in LAN network system. There are some steps that to be followed to crimp UTP cable into RJ 45 connector:

Step 1: Initially cut the plastic sheath about 1 inch from end of cut cable with the use of razor blade located in crimping tool.

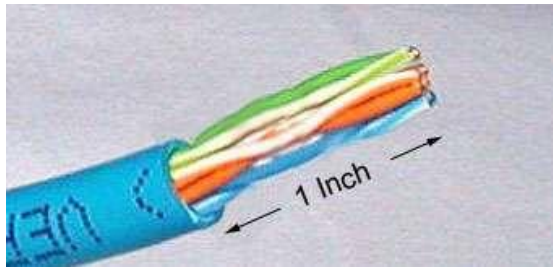


Fig. Cutting of Cable sheath

Step 2: Now unwind and pair the similar colors as shown in fig.

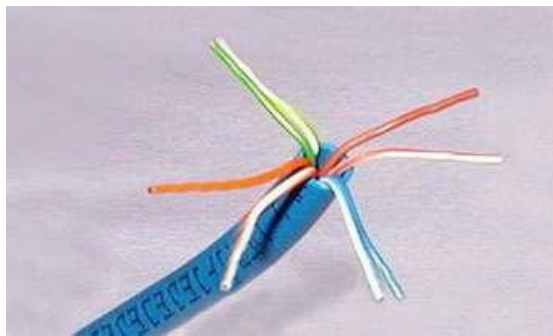


Fig. Unwind the coloured wires

Step 3: Keep the wires in between the fingers and make the wires straight with required colour orders.

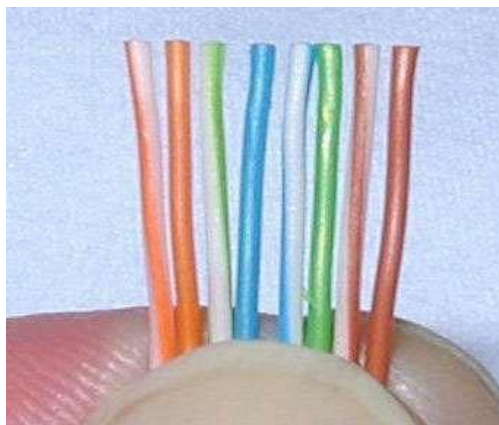


Fig. Straighten wires with fingers

Step 4: Use scissors to make a straight cut across 8 wires to make them short by 1/2 Inch from cut sleeve to wire end.



Fig. Scissor

Step 5: Now insert 8 colour wires in RJ 45 connector by considering proper position of blue plastic sleeve.



Fig. Wires in RJ 45 Connector

Step 6: Now crimp the cable carefully by putting RJ 45 connector in Ethernet Crimper and push on handles tightly. You will find that copper splicing tabs on connector will cut into each of eight wires.



Fig. Crimping of wire

After crimping UTP cable in RJ 45 connector, you find the cable as shown in fig 1.13.

Creating a Switched Wired Ethernet LAN

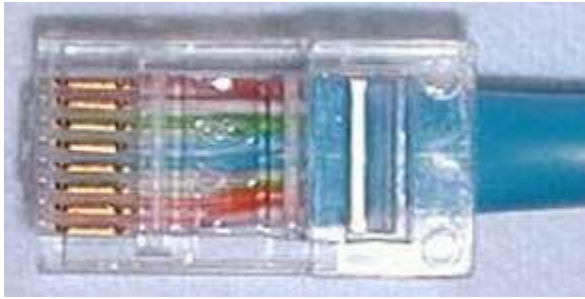


Fig. Cable in RJ 45 Connector

Client and Server

The terms "client" and "server" are used to describe individual computers that are part of a network where computing resources and workload are shared.

A server is a computer that makes its resources available to the network and responds to the commands of a client.

The server's shared resources can be files (a file server); printers (a print server); processing power (an application server); etc...

A client is a computer that uses the resources made available by a server.

The client must have sufficient processing power on its own to run applications that interact with the resources on the server.

It is possible, and quite common, for an individual computer to function as both a client and a server. For example, if Adamu queries a SQL Server database from his workstation for the data he needs to create an Excel spreadsheet, then his workstation is functioning as a client.

However, if Amina then connects to Adamu's workstation from her computer and copies the spreadsheet, then Adamu's workstation is functioning as a server.

OPEN SYSTEMS INTER CONNECTION REFERENCE MODEL (OSI)

Introduction to OSI Model & its layers

The Open Systems Interconnection (OSI) Model was developed by International Organization for Standardization (ISO).

- ISO is the organization, OSI is the model
- It was developed to allow systems with different platforms to communicate with each other. Platform could mean hardware, software or operating system.
- It is a network model that defines the protocols for network communications.
- It is a hierarchical model that groups its processes into layers. It has 7 layers as follows: (Top to Bottom)
 1. Application Layer
 2. Presentation Layer
 3. Session Layer

4. Transport Layer
5. Network Layer
6. Data Link Layer
7. Physical Layer

- Each layer has specific duties to perform and has to cooperate with the layers above and below it.

3.1 Layered Architecture of OSI Model

- The OSI model has 7 layers each with its own dedicated task.
- A message sent from Device A to Device B passes has to pass through all layers at A from top to bottom then all layers at B from bottom to top as shown in the figure below.
- At Device A, the message is sent from the top layer i.e. Application Layer A then all the layers till it reaches its physical layer and then it is transmitted through the transmission medium.
- At Device B, the message received by the physical layer passes through all its other layers and moves upwards till it reaches its Application Layer.

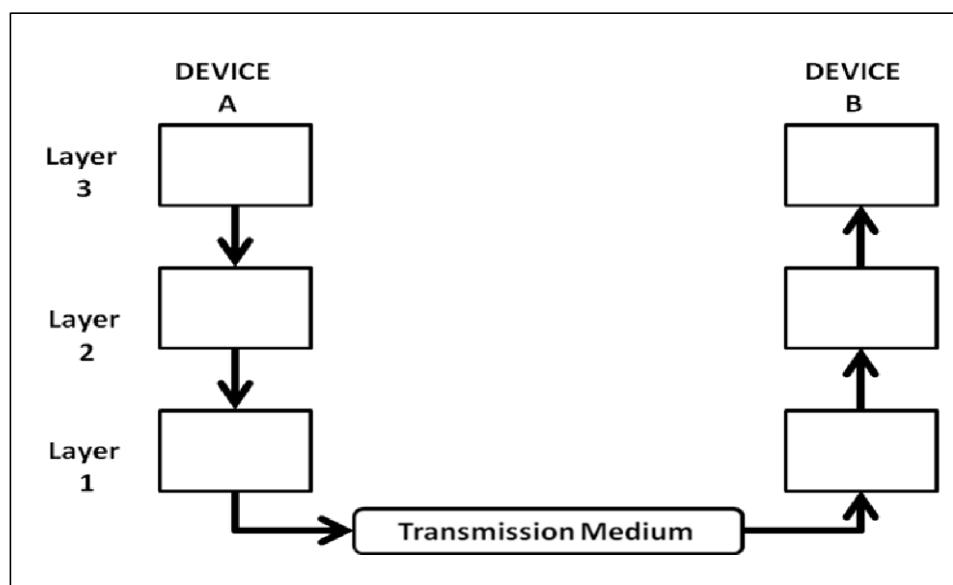


Fig: Flow of Data from Device A to Device B through various layers

As the message travels from device A to device B, it may pass through many intermediate nodes. These intermediate nodes usually involve only the first three layers of the OSI model as shown below.

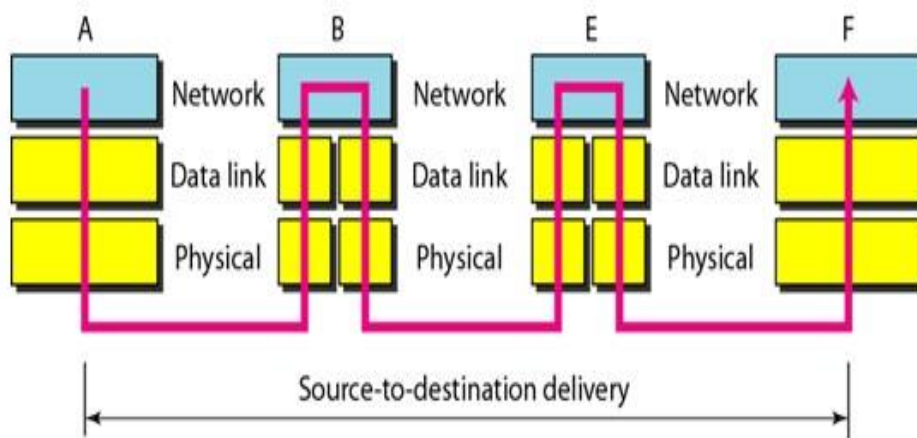


Fig: Data Transfer through Intermediate nodes

The Data Link layer determines the next node where the message is supposed to be forwarded and the network layer determines the final recipient.

3.2 Communication & Interfaces

For communication to occur, each layer in the sending device adds its own information to the message it receives from the layer just above it and passes the whole package to the layer just below it. Each layer in the receiving device removes the information added at the corresponding layer and sends the obtained data to the layer above it.

- Every Layer has its own dedicated function or services and is different from the function of the other layers.
- On every sending device, each layer calls upon the service offered by the layer below it.
- On every receiving device, each layer calls upon the service offered by the layer above it.
- Between two devices, the layers at corresponding levels communicate with each other .i.e. layer 2 at receiving end can communicate and understand data from layer 2 of sending end. This is called peer –to – peer communication.
- For this communication to be possible between every two adjacent layers there is an interface. An interface defines the service that a layer must provide. Every layer has an interface to the layer above and below it as shown in the figure below

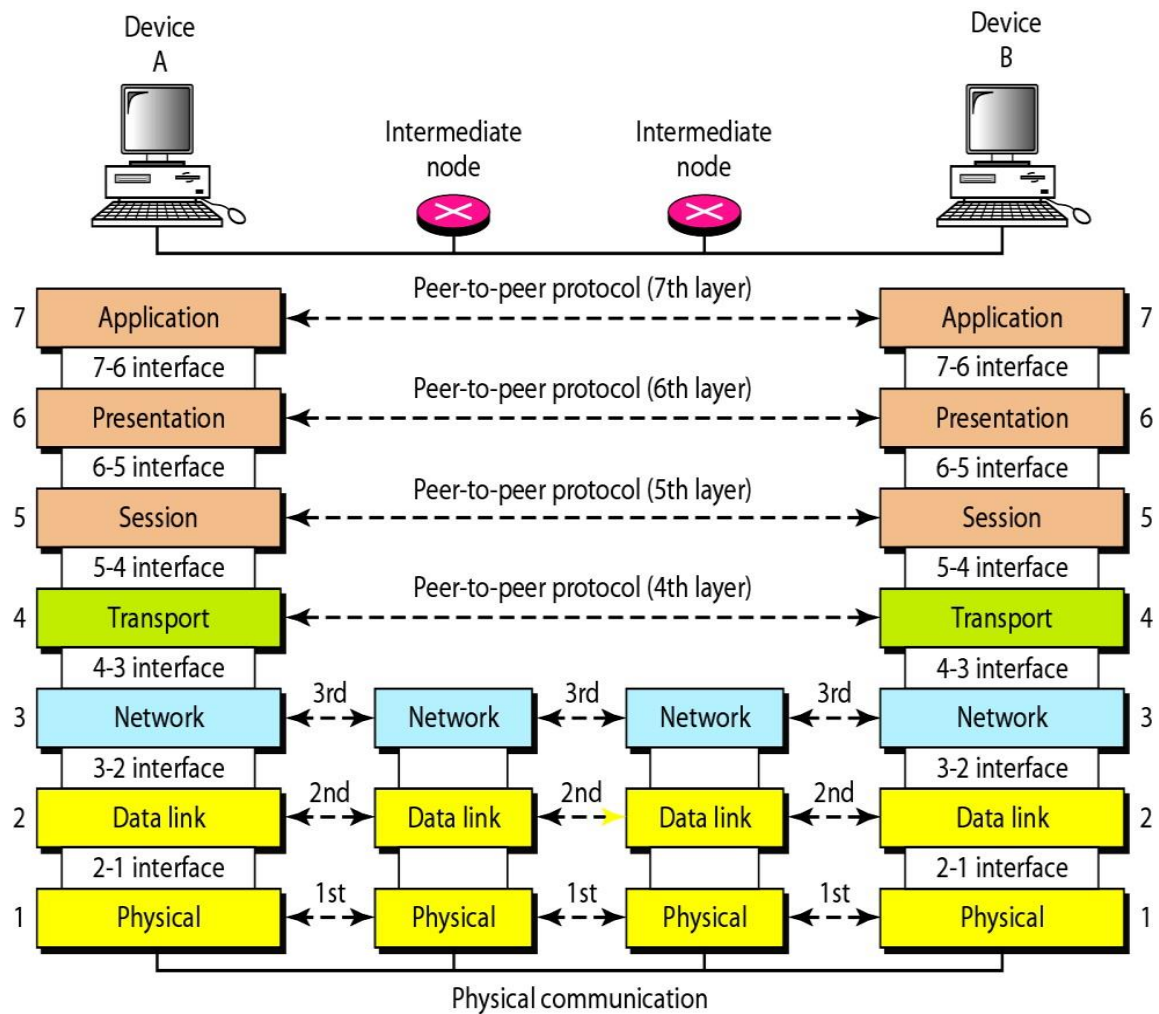


Fig: Communication & Interfaces in the OSI model

Encapsulation of Data

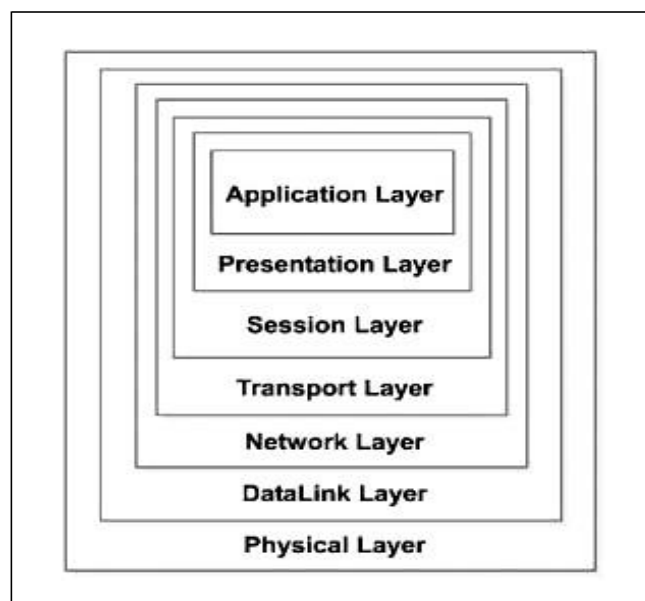


Fig: Encapsulation

As shown in the figure above the data at layer 7 i.e. The Application layer along with the header added at layer 7 is given to layer 6, the Presentation layer. This layer adds its header and passed the whole package to the layer below.

The corresponding layers at the receiving side removes the corresponding header added at that layer and sends the remaining data to the above layer.

The above process is called encapsulation

3.3 Description of Layers in the OSI Model

Physical Layer

1. The Physical Layer provides a standardized interface to physical transmission media, including:

- Mechanical specification of electrical connectors and cables, for example maximum cable length
- Electrical specification of transmission line c. Bit-by-bit or symbol-by-symbol delivery

2. On the sender side, the physical layer receives the data from Data Link Layer and encodes it into signals to be transmitted onto the medium. On the receiver side, the physical layer receives the signals from the transmission medium decodes it back into data and sends it to the Data Link Layer as shown in the figure below:

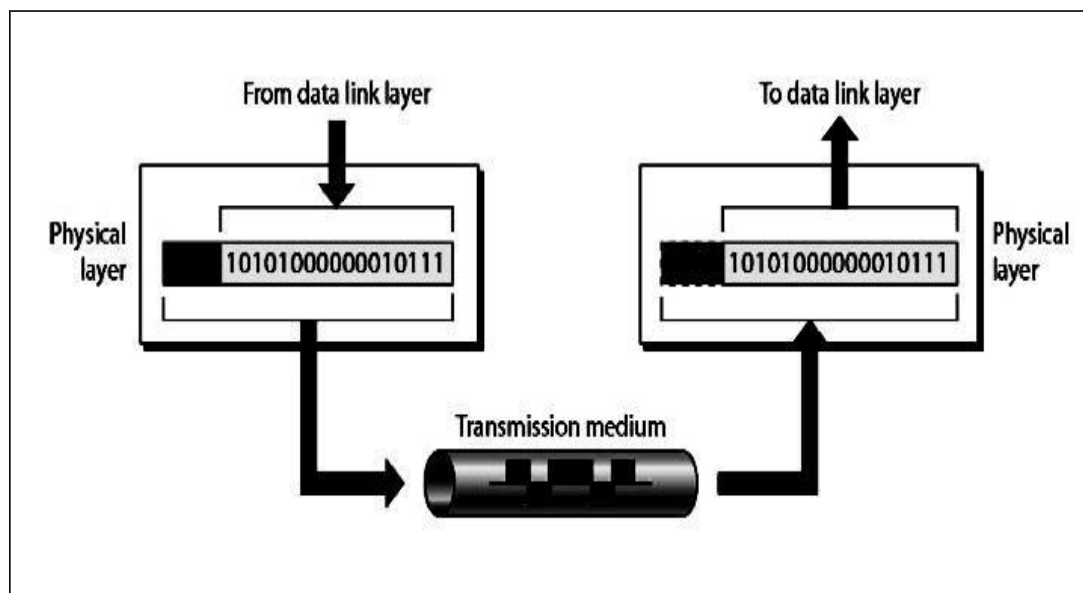


Fig: Transmission of data to and from Physical Layer

3. Interface

The Physical Layer defines the characteristics of interfaces between the devices & transmission medium.

4. **Representation of bits**

The physical layer is concerned with transmission of signals from one device to another which involves converting data (1's & 0's) into signals and vice versa. It is not concerned with the meaning or interpretation of bits.

5. **Data rate**

The physical layer defines the data transmission rate i.e. number of bits sent per second. It is the responsibility of the physical layer to maintain the defined data rate.

6. **Synchronization of bits**

To interpret correct and accurate data the sender and receiver have to maintain the same bit rate and also have synchronized clocks.

7. **Line configuration**

The physical layer defines the nature of the connection .i.e. a point to point link, or a multi-point link.

8. **Physical Topology**

The physical layer defines the type of topology in which the device is connected to the network. In a mesh topology it uses a multipoint connection and other topologies it uses a point to point connection to send data.

9. **Transmission mode**

The physical layer defines the direction of data transfer between the sender and receiver. Two devices can transfer the data in simplex, half duplex or full duplex mode

10. **Main responsibility of the physical layer**

Transmission of bits from one hop to the next.

Data Link Layer

- i. The Data Link layer adds reliability to the physical layer by providing error detection and correction mechanisms.
- ii. On the sender side, the Data Link layer receives the data from Network Layer and divides the stream of bits into fixed size manageable units called as **Frames** and sends it to the physical layer. On the receiver side, the data link layer receives the stream of bits from the physical layer and regroups them into frames and sends them to the Network layer. This process is called **Framing**. It is shown in the figure below:

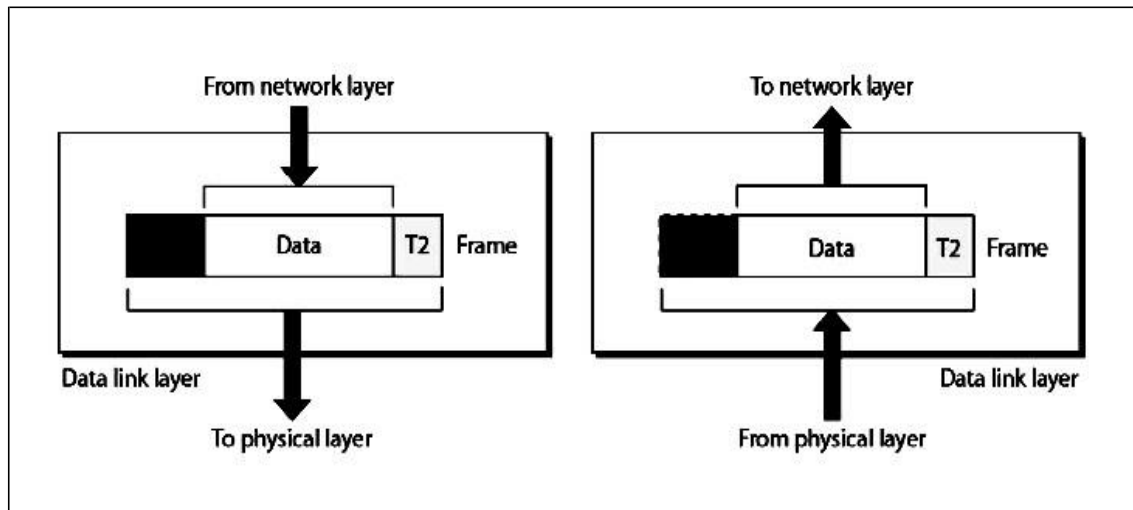


Fig: Data Link Layer: The process of Framing

iii. Physical Addressing (inside / outside sender's network)

- a. The Data link layer appends the physical address in the header of the frame before sending it to physical layer.
- b. The physical address contains the address of the sender and receiver.
- c. In case the receiver happens to be on the same physical network as the sender; the receiver is at only one hop from the sender and the receiver address contains the receiver's physical address.
- d. In case the receiver is not directly connected to the sender, the physical address is the address of the next node where the data is supposed to be delivered.

IV. Flow control

- a. The data link layer makes sure that the sender sends the data at a speed at which the receiver can receive it else if there is an overflow at the receiver side the data will be lost.
- b. The data link layer imposes flow control mechanism over the sender and receiver to avoid overwhelming of the receiver.

V. Error control

- a. The data link layer imposes error control mechanism to identify lost or damaged frames, duplicate frames and then retransmit them.
- b. Error control information is present in the trailer of a frame.

VI. Access Control

- a. The data link layer imposes access control mechanism to determine which device has right to send data in an multipoint connection scenario.

VII. Main Responsibility

- i. The main responsibility of the data link layer is hop to hop transmission of frames.

Network Layer

- i. The network layer makes sure that the data is delivered to the receiver despite multiple intermediate devices.
- ii. The network layer at the sending side accepts data from the transport layer, divides it into packets, adds addressing information in the header and passes it to the data link layer. At the receiving end the network layer receives the frames sent by data link layer, converts them back into packets, verifies the physical address (verifies if the receiver address matches with its own address) and then send the packets to the transport layer.

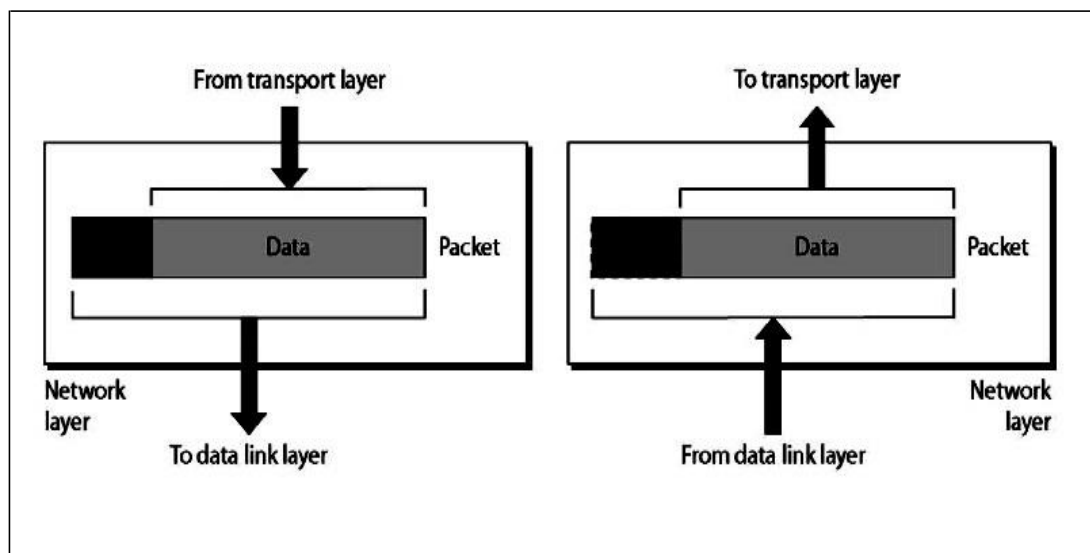


Fig: Network Layer

- iii. The network layer is responsible for source to destination of delivery of data. Hence it may have to route the data through multiple networks via multiple intermediate devices. In order to achieve this network layer relies on two things:

a. Logical Addressing b. Routing

a. Logical Addressing

- The network layer uses logical address commonly known as IP address to recognize devices on the network.

- An IP address is a universally unique address which enables the network layer to identify devices outside the sender's network.
- The header appended by the network layer contains the actual sender and receiver IP address.
 - At every hop the network layer of the intermediate node check the IP address in the header, if its own IP address does not match with the IP address of the receiver found in the header, the intermediate node concludes that it is not the final node but an intermediate node and passes the packet to the data link layer where the data is forwarded to the next node.

b. Routing

The network layer divides data into units called packets of equal size and bears a sequence number for rearranging on the receiving end.

Each packet is independent of the other and may travel using different routes to reach the receiver hence may arrive out of turn at the receiver.

Hence every intermediate node which encounters a packet tries to compute the best possible path for the packet. The best possible path may depend on several factors such as congestion, number of hops, etc.

This process of finding the best path is called as Routing. It is done using routing algorithms.

vi. The Network layer does not perform any flow control or error control

Main Responsibility

The main responsibility of Network Layer is transmission of packets from source to destination

Transport Layer

i. A logical address at network layer facilitates the transmission of data from source to destination device. But the source and the destination both may be having multiple processes communicating with each other. Hence it is important to deliver the data not only from the sender to the receiver but from the correct process on the sender to the correct process on the receiver. The transport layer takes care of process to process delivery of data and makes sure that it is intact and in order.

ii. At the sending side, the transport layer receives data from the session layer, divides it into units called segments and sends it to the network layer. At the receiving side, the transport layer receives packets from the network layer, converts and arranges into proper sequence of segments and sends it to the session layer.

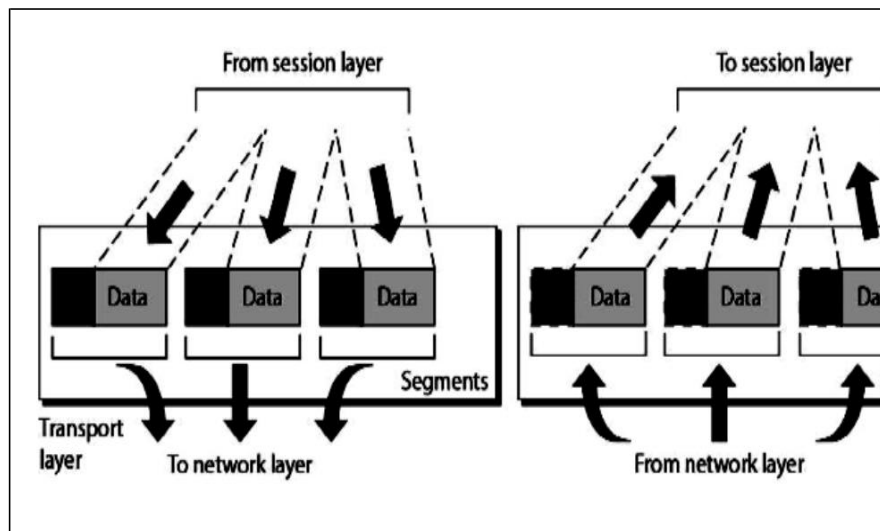


Fig: Transport Layer

iii. To ensure process to process delivery the transport layer makes use of **port address** to identify the data from the sending and receiving process. A Port Address is the name or label given to a process. It is a 16 bit address. Ex. TELNET uses port address 23, HTTP uses port address 80. Port address is also called as Service Point Address

iv. The data can be transported in a connection oriented or connectionless manner. If the connection is connection oriented then all segments are received in order else they are independent of each other and are received out of order and have to be rearranged.

v. The Transport layer is responsible for segmentation and reassembly of the message into segments which bear sequence numbers. This numbering enables the receiving transport layer to rearrange the segments in proper order.

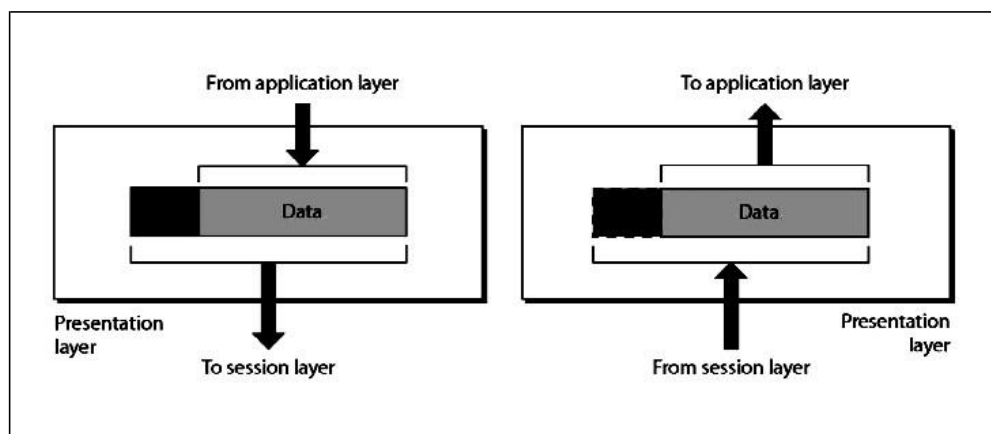
vi. Flow Control & Error control: the transport layer also carries out flow control and error control functions; but unlike data link layer these are end to end rather than node to node.

vii. Main Responsibility

The main responsibility of the transport layer is process to process delivery of the entire message.

Session Layer

- i. The session layer establishes a session between the communicating devices called dialog and synchronizes their interaction. It is the responsibility of the session layer to establish and synchronize the dialogs. It is also called the network dialog controller.
- ii. The session layer at the sending side accepts data from the presentation layer adds checkpoints to it called syn bits and passes the data to the transport layer. At the receiving end the session layer receives data from the transport layer removes the checkpoints inserted previously and passes the data to the presentation layer.
- iii. The checkpoints or synchronization points is a way of informing the status of the data transfer. Example A checkpoint after first 500 bits of data will ensure that those 500 bits are not sent again in case of retransmission at 650th bit.



- vi. Main responsibility of session layer is dialog control and synchronization

Presentation Layer

- i. The communicating devices may be having different platforms. The presentation layer performs translation, encryption and compression of data.

ii. The presentation layer at sending side receives the data from the application layer adds header which contains information related to encryption and compression and sends it to the session layer. At the receiving side, the presentation layer receives data from the session layer decompresses and decrypts the data as required and translates it back as per the encoding scheme used at the receiver.

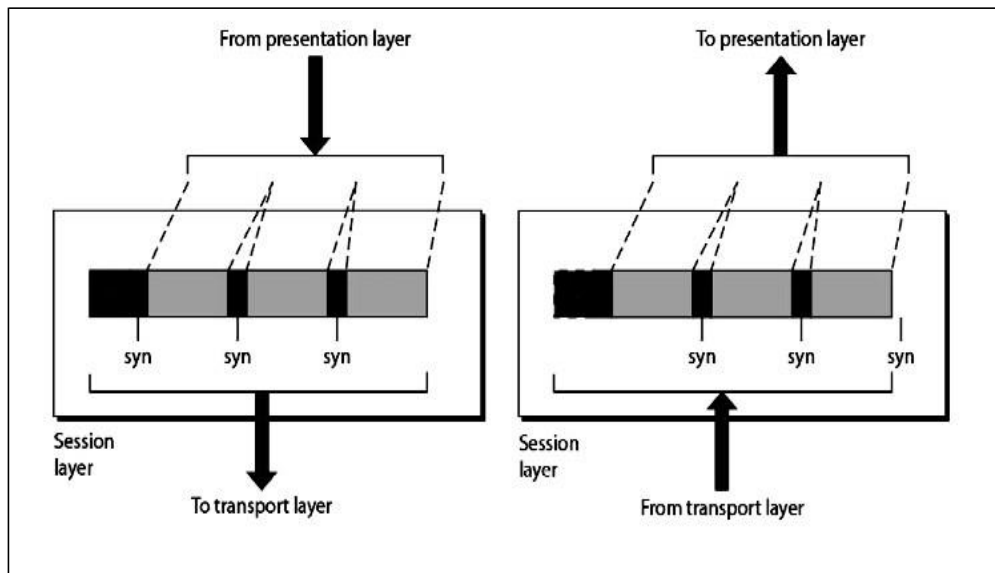


Fig : Presentation Layer

iii. Translation

The sending and receiving devices may run on different platforms (hardware, software and operating system). Hence it is important that they understand the messages that are used for communicating. Hence a translation service may be required which is provided by the Presentation layers

iv. Compression

Compression ensures faster data transfer. The data compressed at sender has to be decompressed at the receiving end, both performed by the Presentation layer.

v. Encryption

It is the process of transforming the original message to change its meaning before sending it. The reverse process called decryption has to be performed at the receiving end to recover the original message from the encrypted message.

vi. Main responsibility

The main responsibility of the Presentation layer is translation, compression and encryption.

Application Layer

- i. The application layer enables the user to communicate its data to the receiver by providing certain services. For ex. Email is sent using X.400 service.

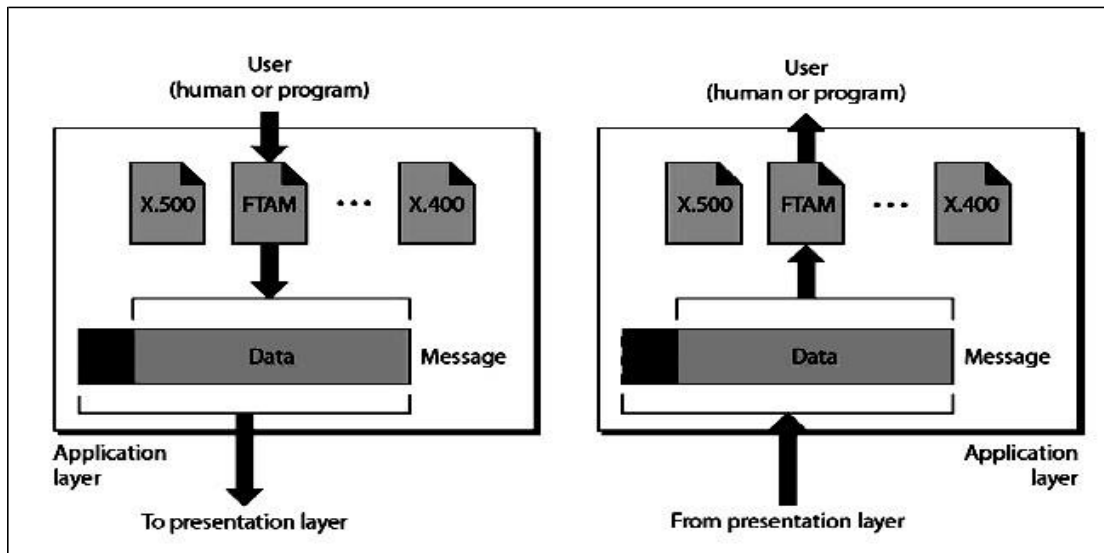


Fig : Application Layer

- ii. **X500** is a directory service used to provide information and access to distributed objects
- iii. **X400** is services that provides basis for mail storage and forwarding
- iv. **FTAM (File transfer, access and management)** provides access to files stored on remote computers and mechanism for transfer and manage them locally.
- v. Main Responsibility

Main Responsibility of Application layer is to provide access to network resources.

SUMMARY

The responsibilities of the 7 layers of OSI model can be summarized as follows:

1. Application Layer : To provide the users access to network resources
2. Presentation Layer: To provide the functions of translation, encryption and compression.
3. Session Layer: To establish, manage and terminate sessions
4. Transport Layer: To provide process to process delivery of message
5. Network Layer: To provide source to destination delivery of packets.
6. Data link Layer: To provide hop to hop delivery of frames

7. Physical Layer: To transmit data over a bit stream from one hop to the next and provide electrical and mechanical specification.

Difference between IPv4 and IPv6

IPv4 and IPv6 are internet protocol versions, with IPv6 being an upgraded version of IPv4. There are several differences between the IPv4 and IPv6 protocols, including their functionality, but the most important difference is the quantity of addresses (Address space) that they create.

Read through this article to find out more about IPv4 and IPv6 and how they are different from each other.

What is Internet Protocol (IP)?

The Internet Protocol is a set of rules that allows our computers to communicate via the Internet. IP addresses are basically in charge of directing the data packets to their correct destinations. IP controls all Internet traffic. Data packets containing the IP addresses of their points of origin and destinations travel on the Internet.

When you type an URL on your browser, a data packet including your IP address is transmitted to the web server's IP address, and the website corresponding to the URL is then served back to your device over the Internet.

What is IPv4?

IPv4 is a major protocol in the TCP/IP suite. IPv4 addresses provide a way to uniquely identify the hosts in a network. IPv4 uses 32-bit logical addresses. Depending on the network type, there are various ways to configure IPv4 with multiple devices, including manual and automatic configurations.

IPv4 uses the best-effort model, which means it does not guarantee delivery of data to a host or avoid duplicate delivery.

For Ethernet communication, IPv4 uses five classes of 32-bit addresses – A, B, C, D, and E. The bit length for addressing the network host differs between Classes A, B, and C. Multicasting is reserved for class D addresses, whereas future use is reserved for class E addresses. IPv4 has a limit of 4.20×10^9 host addresses that it can allocate to end-users.

What is IPv6?

As the popularity of the Internet grew in the 1990s, it was soon realized that the number of IP addresses that IPv4 can create is restricted, and it will get exhausted. Hence, the Internet

Engineering Task Force (IETF) had developed IPv6, which is a better and upgraded version of IPv4 that was intended to replace IPv4 eventually.

IPv6 provides 128-bit IP addresses, which means it can generate 3.4×10^{38} addresses. In layman's terms, it can generate trillions of trillions IP addresses. IPv6 reserves blocks of numbers for particular purposes. It prohibits using specific numbers entirely, so the total number of IPv6 addresses should be slightly less. Nonetheless, the amount of IPv6 addresses is almost endless, so there would be no dearth of IP addresses in the future.

IPv6 addresses follow the same design principles. The addresses are divided into eight groups of four hexadecimal digits separated by colons. Most IPv6 addresses do not use all of their 128 bits, resulting in fields that are either entirely zeros or are padded with zeros. The two colons (::) can represent a contiguous 16-bit field of zeros in IPv6 addressing scheme.

Difference between IPv4 and IPv6

The following table highlights the major differences between IPv4 and IPv6 –Key IPv4 Protocol

IPv6 Protocol

Address Configuration	IPv4 has an address length of 32-bit represented in decimal format. It supports Manual and DHCP configuration.	IPv6 has a 128-bit address length represented in hexadecimal format. It supports auto-configuration and renumbering configuration.
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Address Space In case of IPv4, a total number of 4.20×10^9 addresses could get generated.

In case of IPv6, the total number of IP addresses that could be generated is 3.4×10^{38} which is much greater than the number produced by IPv4.

Secure IPv4 is being used as less secure protocol, as its security section is dependent on the application, i.e., it is proportional to the security that is provided or implemented at the application level. IPv6 has its inbuilt security feature named as IPSEC (Internet Protocol Security) which provide additional security feature along with the security provided or implemented at application level.

Fragmentation In case of IPv4, fragmentation is performed by both the Sender and the Forwarding routers. In case of IPv6, fragmentation is performed only by the sender routers.

Authentication IPv4 does not provide Encryption and Authentication facilities. IPv6 provides both Encryption and Authentication facilities.

Header Size In IPv4, the Request header is not fixed and it may be between of 20-60 bytes in size. In IPv6, the Request header is of fixed size, i.e., 40 bytes.

Conclusion

IPv4 creates 4.20×10^9 unique network addresses, which is an insufficient in number, and as a result, the Internet is running out of space. IP version 6 (IPv6), on the other hand, generates 3.4×10^{38} addresses and provides a scalable and adaptable solution to the existing issue.